GENETICS

Genetic Disorders of the Urinary Tract

LG 1 · Assist. Prof. Dr. Evan Latef Khlef
GUE BLOCK

Genetics GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which two embryonic tissues build the kidney, and which part does each one form?
2. What is the typical recurrence risk per pregnancy for an autosomal dominant, an autosomal recessive and an X-linked kidney disorder?
3. A child has enlarged echogenic kidneys and congenital hepatic fibrosis. Which disease is this, and which gene?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

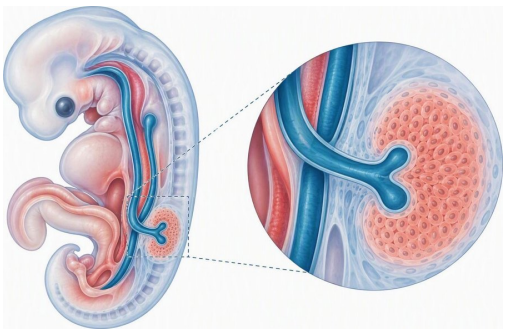
1 From gene to kidney disease — six steps

A variant can break the kidney at any stage, from forming it to running its cells. Learn the disorder that sits at each step.

Step	Biological process	Disorder	Associated genes
1 Kidney formation	Induction and tissue formation	**Renal agenesis**; renal dysplasia	GREB1L, HNF1B and others
2 Kidney growth	Branching and nephron formation	**Renal hypoplasia**; renal dysplasia	Several possible — depends on the phenotype
3 Urine drainage	Urinary tract connection	**VUR**; congenital obstruction	Multiple genes may contribute
4 Specialised cell function	Glomerular structure and **filtration barrier**	**Alport syndrome**	COL4A3, COL4A4, COL4A5
5 Tubular and ciliary function	Tubular integrity and cell signalling	**Nephronophthisis** (NPHP)	NPHP1 and other NPHP genes
6 Cellular signalling	**Polycystin signalling** and cyst formation	**ADPKD**	PKD1, PKD2

2 Embryology — two tissues, one organ

Reciprocal induction — each tissue makes the other develop	
4–5 weeks	**Ureteric bud** sprouts from the mesonephric duct
5–10 weeks	**Branching morphogenesis** + nephron induction
~34–36 weeks	Nephrogenesis largely **completes before birth**
Collecting system	← **ureteric bud**
Nephrons	← **metanephric mesenchyme**



Lecture figure — the ureteric bud (blue) branching into the metanephric mesenchyme (pink).

Five regulatory nodes of urinary tract development	
WT1	Supports kidney development and differentiation
GDNF → RET	Promotes **ureteric bud formation and outgrowth**
PAX2	Supports kidney and urinary tract development
HNF1B	Patterns tubules and **multiple organs**
Branching signals	Balance growth and differentiation

Genes are pathway markers — not one-gene / one-malformation labels.

3 CAKUT — a spectrum, not one diagnosis

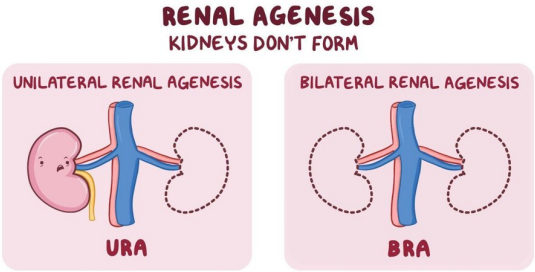
Diagnostic yield	
12–20%	may have an identified **monogenic cause**
4–11%	may have a pathogenic **copy-number variant**
Why it varies	Phenotyping depth · **syndromic clues** · family history · test platform

The four groups	
Number / position	**Agenesis** · ectopia · horseshoe kidney
Parenchyma	**Hypoplasia** · dysplasia · cystic dysplasia
Collecting system	**Duplex system** · hydronephrosis · UPJ obstruction
Outflow / reflux	**Posterior urethral valves** · VUR

Yields are cohort- and method-dependent.

4 Renal agenesis — incidental to life-limiting

	Unilateral	Bilateral
Picture	**Often asymptomatic**	**Severe fetal phenotype**
What happens	**Compensatory hypertrophy** of the solitary kidney	Anuria → **oligohydramnios** → pulmonary hypoplasia
Look for	Contralateral CAKUT · **genital tract anomalies** · hypertension / albuminuria over time	Prenatal: absent kidneys / renal arteries · **non-visualised bladder** · reduced amniotic fluid
Counsel	**Protect the solitary kidney**; screen relatives selectively	Confirm anatomy; discuss prognosis, **recurrence** and genomic evaluation

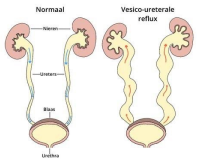


Lecture figure — unilateral (URA) vs bilateral (BRA) renal agenesis.

5 Vesicoureteral reflux (VUR) — anatomy → infection → scarring

Short intramural ureteric tunnel → **Retrograde urine flow** → **Febrile UTI / pyelonephritis** → **Scarring risk in susceptible kidneys**

Management	**Risk-based** — age · febrile UTIs · grade · bladder–bowel dysfunction · renal scars · adherence
Genetics	**Familial aggregation is common**, but most isolated VUR is genetically complex
On the gene map	Step 3 (urine drainage) — multiple genes may contribute



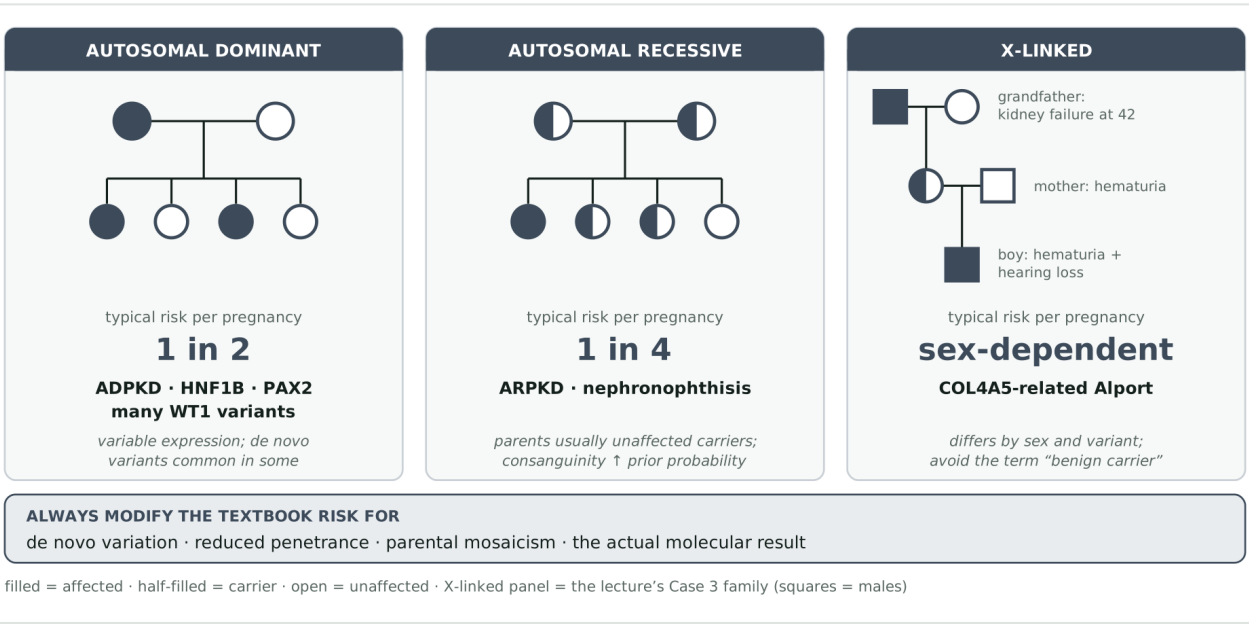
Lecture figure — normal (left) vs reflux (right). Labels are Dutch: nieren = kidneys, blaas = bladder.

6 Genetic testing — a phenotype-first ladder

1 Phenotype	Imaging, urine, hearing, eyes, liver, glucose, magnesium, **pedigree**
2 Copy number	**Chromosomal microarray** or validated CNV calling when CAKUT is **syndromic**
3 Sequence	**Focused panel** when the phenotype is specific; **exome/genome** when heterogeneous
4 Interpret	ACMG classification + phenotype fit + segregation — **a VUS is not a diagnosis**
5 Act	Translate the result into surveillance, treatment, **family testing** and counselling

> **Negative ≠ non-genetic.** Reanalysis matters as genes, classifications and phenotypes evolve. Use a laboratory method that **detects the variant classes** relevant to the suspected disorder.

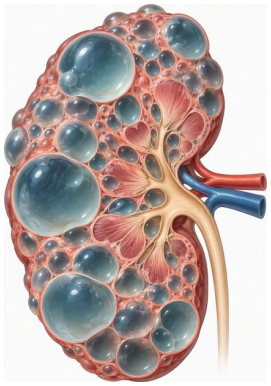
7 Inheritance changes recurrence risk



Inheritance schema built from the lecture's inheritance slide; the X-linked family is the lecture's Case 3.

8 Polycystic kidney disease — ADPKD vs ARPKD

ADPKD — SYSTEMIC AND AGE-DEPENDENT	
Genes	**PKD1, PKD2** — autosomal dominant; cyst burden increases with age
Kidney	**Bilateral cysts**, hypertension, pain, hematuria, stones, progressive CKD
Beyond kidney	**Liver cysts**; selected patients need **intracranial aneurysm** risk assessment
Diagnosis	**Age-specific imaging criteria**; genetic testing when imaging or family history is equivocal, or for reproductive / donor decisions
Management	**Blood-pressure control**, CKD care, lifestyle; disease-modifying therapy for appropriate rapidly progressive disease



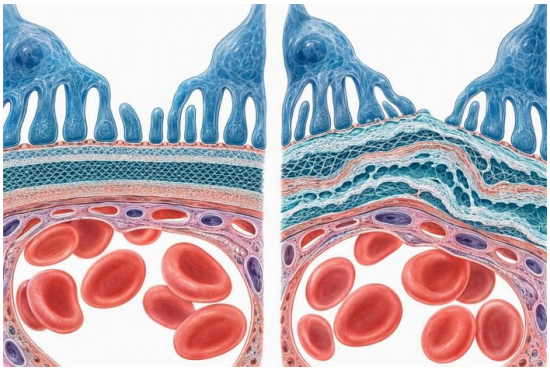
Lecture figure — ADPKD kidney filled with variable-sized cysts.

ARPKD — COLLECTING-DUCT DISEASE WITH HEPATIC FIBROSIS	
Gene	**PKHD1** — autosomal recessive; carrier × carrier → **25% affected** per pregnancy
Kidney	**Enlarged, echogenic kidneys**; dilated collecting ducts · neonatal respiratory compromise · childhood hypertension · progressive CKD
Liver	**Congenital hepatic fibrosis** — portal hypertension · splenomegaly · varices · recurrent cholangitis in some
Timeline	Prenatal ultrasound may show enlarged echogenic kidneys and **reduced amniotic fluid**; later, **liver disease may dominate** even when renal function is relatively preserved

They separate by pattern — not cysts alone	ADPKD	ARPKD
Inheritance	Autosomal dominant	Autosomal recessive
Typical timing	**Adolescence / adulthood**; age-dependent	**Prenatal, neonatal or childhood**
Kidney pattern	Bilateral **variable-sized cysts**; enlarged kidneys	Enlarged **echogenic** kidneys; collecting-duct ectasia
Liver pattern	**Cysts** / polycystic liver disease	**Congenital hepatic fibrosis** / portal hypertension
Genes / test	PKD1, PKD2; consider phenocopies	PKHD1; broad cystic panel if atypical

9 Alport syndrome — a collagen IV basement-membrane disorder

Genes	COL4A3 · COL4A4 · COL4A5
Defect	Abnormal **α3α4α5 type IV collagen**
Kidney	**Persistent hematuria** → proteinuria → CKD
Ear	**Sensorineural hearing loss**
Eye	**Lenticonus** and characteristic retinal changes in some patients
Inheritance	**X-linked is common** (COL4A5-related); autosomal recessive and autosomal dominant forms also occur



Lecture figure — glomerular filtration barrier: normal (left) vs the abnormal basement membrane (right).

10 The lecture's three cases

Case	Clues	Steps	Take-home
1 Newborn with oligohydramnios	Term infant; **absent left kidney**; small echogenic right kidney; no other obvious anomaly	**Stabilise + phenotype** (urine output, creatinine trend, ultrasound, BP) → **look beyond the kidney** (genitals, spine, heart, hearing, eyes; prenatal history) → **choose testing** (microarray + CAKUT panel / exome) → **interpret in family** (parental imaging, segregation)	**Test the infant** — not only the parents — and pair sequence analysis with copy-number detection
2 Age 28, hypertension + family history	BP 152/94; eGFR normal; **bilateral renal cysts**; father reached kidney failure at 54	**Confirm** (imaging pattern, family history) → **stratify** (kidney size, eGFR trend, BP, complications) → **test?** → **counsel** → **act** (treat BP; assess progression risk)	Testing is useful if atypical, family variant unknown, or planning; **50% transmission**; respect the right not to know
3 Boy of 14, persistent hematuria	Microscopic hematuria since age 6; new albuminuria; normal complement; **high-frequency hearing loss**; mother: hematuria; maternal grandfather: kidney failure at 42	Suspect **X-linked Alport** → test **COL4A3/4/5 sequence + deletion/duplication** → act (nephrology care, hearing/eye assessment, early kidney-protective therapy) → **cascade** (targeted testing and surveillance for at-risk relatives)	**Persistent hematuria** is the diagnostic clue

P Must-remember pearls

- **Ureteric bud → collecting system; metanephric mesenchyme → nephrons.** They induce each other; GDNF → RET drives ureteric bud outgrowth.
- **Bilateral renal agenesis:** anuria → oligohydramnios → pulmonary hypoplasia. Unilateral is often asymptomatic, with compensatory hypertrophy of the other kidney.
- **AD 1 in 2** (ADPKD, HNF1B, PAX2) · **AR 1 in 4** (ARPKD, nephronophthisis) · **X-linked sex-dependent** (COL4A5 Alport). Always adjust for de novo variants, reduced penetrance and parental mosaicism.
- **ADPKD** (PKD1/PKD2): adult, variable-sized cysts, liver cysts, aneurysm assessment in selected patients. **ARPKD** (PKHD1): prenatal or childhood, enlarged echogenic kidneys, congenital hepatic fibrosis.
- **Alport** = abnormal α3α4α5 type IV collagen: persistent hematuria → proteinuria → CKD, plus sensorineural hearing loss and lenticonus. In testing, a **VUS is not a diagnosis** and negative ≠ non-genetic.

Q Test yourself

1. A newborn struggles to breathe. Prenatal scans showed no kidneys, a non-visualised bladder and almost no amniotic fluid. What explains the breathing problem?

A) Congenital hepatic fibrosis
B) Pulmonary hypoplasia from oligohydramnios
C) Pyelonephritis from reflux
D) Compensatory hypertrophy

2. Two healthy parents have a child with ARPKD. What is the risk for each future pregnancy?

A) 1 in 2
B) Sex-dependent
C) Nil
D) 1 in 4

3. A 30-year-old has bilateral variable-sized kidney cysts and liver cysts. His father reached kidney failure at 50. Which genes?

A) PKHD1
B) COL4A5
C) PKD1 or PKD2
D) NPHP1

4. A boy has persistent microscopic hematuria and sensorineural hearing loss. Which protein is abnormal?

A) α3α4α5 type IV collagen
B) Polycystin
C) The GDNF → RET signal
D) HNF1B

5. An exome for a child with renal dysplasia reports a variant of uncertain significance (VUS). What does it mean?

A) The diagnosis is confirmed
B) It is not a diagnosis — weigh ACMG class, phenotype fit, segregation
C) The cause is not genetic
D) The parents need no testing

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1. B — Pulmonary hypoplasia from oligohydramnios — bilateral agenesis: anuria → oligohydramnios → pulmonary hypoplasia.

2. D — 1 in 4 — both parents are carriers of a PKHD1 variant; 25% affected per pregnancy.

3. C — PKD1 or PKD2 — adult onset, liver cysts and a dominant family history point to ADPKD.

4. A — α3α4α5 type IV collagen — Alport syndrome — a collagen IV basement-membrane disorder.

5. B — It is not a diagnosis — weigh ACMG class, phenotype fit, segregation — a VUS is not a diagnosis; segregation in the family helps clarify it.

SPACED REVIEW

☐ Day 1 | ☐ Day 3 | ☐ Day 7 | ☐ Day 21 | ☐ Pre-exam